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BILATERAL RAMANUJAN SERIES AND THEIR p-ADIC MATES

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Abstract. We conjecture p-adic identities associated to bilateral rational Ramanujan-like series. Then, we show that those identities for $3m+1$ primes imply the corresponding Ramanujan-like series for π^{-m}

1. A note on p-adic series

We compare Taylor expansions with conjectured p-adic expansions obtained replacing x with νp and using p-adic analogues.

Example 1.1. The identity

$$x+x^2+x^3+x^4+\cdots=\frac{x}{1-x} \tag{1}$$

has sense in the reals for $|x|<1$. Replacing x with p (a prime number), we have

$$p+p^2+p^3+p^4+\cdots=\frac{p}{1-p}, \tag{2}$$

which has no sense in the reals, but has sense in the p-adics, that is

$$p+p^2+p^3+p^4+\cdots+p^k+\frac{p}{p-1}\equiv 0 \pmod{p^{k+1}}. \tag{3}$$

for $k\geq 1$.

Example 1.2. From the identity

$$\sum_{n=0}^{\infty}\left(\frac{1}{n+1}-\frac{1}{n+x+1}\right)=\zeta(2)x-\zeta(3)x^2+\zeta(4)x^3-\zeta(5)x^4+\cdots, \tag{4}$$

and after differentiating it, we easily arrive at

$$\sum_{n=0}^{\infty}\frac{x^2}{(n+1)^2}-\sum_{n=0}^{\infty}\frac{x^2}{(n+x+1)^2}=2\zeta(3)x^3-3\zeta(4)x^4+4\zeta(5)x^5-5\zeta(6)x^6+\cdots. \tag{5}$$

Denote

$$S(N)=N^2\sum_{n=0}^{N-1}\frac{1}{(n+1)^2},$$

Replacing x with νp in the left side of (4), we get $S(\nu p)$. Then taking into account that $\zeta_p(\nu)=0$ when ν is an even integer different from 0, we conjecture the following p-adic expansions:

$$S(\nu p)=S(\nu)+2\zeta_p(3)\nu^3p^3+4\zeta_p(5)\nu^5p^5+\cdots.$$

We can check the above p-adic expansion using the following theorem: For integers k and primes $p\geq k+2$, it holds [4, page 10-top]:

$$\zeta_p(k) \equiv \frac{B_{p-k}}{k} = \zeta(1+k-p) \pmod{p}, \tag{6}$$

and for integers k and $n \geq 2$, and primes $p \geq k+2$, it holds [4, page 10-top]:

$$\zeta_p(k) \equiv \frac{B_{p^{n-1}(p-1)+1-k}}{k-1}\left(1-\frac{p^{n-1}}{k-1}\right) \pmod{p^n}. \tag{7}$$

MAPLE PROGRAM.

```
zp:=(n,k,p)->bernoulli(p^(n-1)*(p-1)+1-k)/(k-1)*(1-p^(n-1)/(k-1)) mod p^n:
t:=p->sum(p^2/(n+1)^2,n=0..p-1)-1-2*zp(4,3,p)*p^3-4*zp(4,5,p)*p^5 mod p^7:

t(11),t(13),t(17),t(19),t(23);
```

Example 1.3. From the BBP formula

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}\left(\frac{2}{4n+1}+\frac{2}{4n+2}+\frac{1}{4n+3}\right) = \pi,$$

we have

$$\begin{aligned} &x\sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}\left(\frac{2}{4n+1}+\frac{2}{4n+2}+\frac{1}{4n+3}\right) \\ &-x\sum_{n=0}^{\infty} \frac{(-1)^n}{4^{n+x}}\left(\frac{2}{4(n+x)+1}+\frac{2}{4(n+x)+2}+\frac{1}{4(n+x)+3}\right) = 16L_{-4}(2)x^2+\cdots, \end{aligned} \tag{8}$$

where $L_{-4}(2)$ is the Catalan constant. Let

$$S(N) = N\sum_{n=0}^{N-1} \frac{(-1)^n}{4^n}\left(\frac{2}{4n+1}+\frac{2}{4n+2}+\frac{1}{4n+3}\right).$$

If we replace x with νp in the left hand side of (8), we get $S(\nu p)$. Then, we claim the following p adic identities:

$$S(\nu p) = \left(\frac{\chi}{p}\right) S(\nu) + 16L_{-4,p}(2)\nu^2 p^2 + \cdots.$$

2. Bilateral series of Ramanujan-type

In this paper, we only consider rational Ramanujan-like series. They can be written as

$$\sum_{n=0}^{\infty}\left(\prod_{i=0}^{2m}\frac{(s_i)_n}{(1)_n}\right)\sum_{k=0}^{m} a_k n^k z_0^n = \frac{\sqrt{(-1)^m\chi}}{\pi^m}, \tag{9}$$

where z_0 is a rational, $a_0, a_1, ..., a_m$ are positive rationals, and χ_0 the discriminant of a certain quadratic field (imaginary or real), which is an integer. If $|z_0| > 1$ then the series

on the left side of (9) is divergent, but we understand it as the analytic continuation of the series with z at $z = z_0$. In [3], and using the following definition:

$$R(n,x) = \left(\prod_{i=0}^{2m} \frac{(s_i)_{n+x}}{(1)_{n+x}}\right) \sum_{k=0}^{m} a_k (n+x)^k \, z_0^{n+x}. \tag{10}$$

we proved the bilateral Ramanujan-like series identity

$$\sum_{n=-\infty}^{\infty} R(n,x) - \frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}}$$
$$= \frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}} e^{i\pi x} \prod_{s_k} \frac{1-\cos \pi s_k}{\cos \pi x - \cos \pi s_k} \sum_{k=1}^{m} \left(\alpha_k (\cos 2\pi k x - 1) + \beta_k \sin 2\pi k x\right), \tag{11}$$

which can be written as

$$\sum_{n=0}^{\infty} R(n,x) - \frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}}$$
$$= \frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}} e^{i\pi x} \prod_{s_k} \frac{1-\cos \pi s_k}{\cos \pi x - \cos \pi s_k} \sum_{k=1}^{m} \left(\alpha_k (\cos 2\pi k x - 1) + \beta_k \sin 2\pi k x\right)$$
$$+ \sum_{n=1}^{\infty} R(-n,x), \tag{12}$$

where the α_k and β_k are constants, that we conjecture are rational, see [2], and

$$R(-n,x) = x^{2m+1} \left(\prod_{i=0}^{2m} \frac{(1)_{n-x}}{(s_i)_{n-x}}\right) \sum_{k=0}^{m} a_k (-n+x)^{k-2m-1} \, z_0^{-n} = \mathcal{O}(x^{2m+1}). \tag{13}$$

Developping the last sum (sum to the left side) of (12), we have

$$\frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}} - \sum_{n=0}^{\infty} R(n,x) = \sum_{n=0}^{\infty} R(n,0) - \sum_{n=0}^{\infty} R(n,x)$$
$$= -\frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}} e^{i\pi x} \prod_{s_k} \frac{1-\cos \pi s_k}{\cos \pi x - \cos \pi s_k} \sum_{k=1}^{m} \left(\alpha_k (\cos 2\pi k x - 1) + \beta_k \sin 2\pi k x\right)$$
$$+ (A + Bx + Cx^2 + \cdots) x^{2m+1}, \tag{14}$$

We conjecture that A is of the form $A = rL_\chi(m+1)$, where r is a rational number. Then, we see that

$$\frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}} - \sum_{n=0}^{\infty} R(n,x) = \sum_{n=0}^{\infty} R(n,0) - \sum_{n=0}^{\infty} R(n,x)$$
$$= -\frac{\sqrt{(-1)^m \chi}}{\Gamma(\frac{1}{2})^{2m}} e^{i\pi x} \prod_{s_k} \frac{1-\cos \pi s_k}{\cos \pi x - \cos \pi s_k} \sum_{k=1}^{m} \left(\alpha_k (\cos 2\pi k x - 1) + \beta_k \sin 2\pi k x\right)$$
$$+ rL_\chi(m+1) x^{2m+1} + Bx^{2m+2} + Cx^{2m+3} + \cdots. \tag{15}$$

Denote

$$S(N)=\sum_{n=0}^{N-1}\left(\prod_{i=0}^{2m}\frac{(s_i)_n}{(1)_n}\right)\sum_{k=0}^{m}a_k n^k z_0^n,\qquad T(N)=\left(\prod_{i=0}^{2m}\frac{(s_i)_N}{(1)_N}\right)z_0^N, \tag{16}$$

Observe that if we replace x with νp, $\nu=1,2,3,\cdots$ in the left hand side of (12), we get $S(\nu p)$. Then, replace x with νp in the Fourier part (second line of (12), and observe that we get 0. Finally, replace $L_\chi(m+1)$ with its p-adic analogue $L_{\chi,p}(m+1)$, and the same for $B,C,\cdots$.

Conjecture 2.1. The following p-adic identities hold:

$$S(\nu p)=\left(\frac{\chi}{p}\right)S(\nu)p^m+T(\nu)\left\{rL_{\chi,p}(m+1)\nu^{2m+1}p^{2m+1}+B_p\nu^{2m+2}p^{2m+2}+\cdots\right\}, \tag{17}$$

where $\nu=1,2,3,\ldots$.

Corollary 2.2.

$$S(\nu p)\equiv\left(\frac{\chi}{p}\right)S(\nu)p^m \pmod{\nu^{2m+1}p^{2m+1}}.$$

The case $\nu=1$ implies the following supercongruences (conjectured in [7] by Wadim Zudilin):

$$S(p)\equiv a_0\left(\frac{\chi}{p}\right)p^m \pmod{p^{2m+1}}, \tag{18}$$

and even those conjectured by Y. Zhao up to the rational coefficients [6]. Observe that we give explicit formulas for the coefficients.

3. Examples

Example 3.1. From

$$\sum_{n=0}^{\infty}\frac{\left(\frac12\right)_n^5}{(1)_n^5}\frac{(-1)^n}{4^n}(20n^2+8n+1)=\frac{8}{\pi^2}.$$

We see that

$$R(-n,x)=7\cdot 2^6\zeta(3)x^5+Bx^6+Cx^7+\cdots.$$

Denoting

$$S(N)=\sum_{n=0}^{N-1}\frac{\left(\frac12\right)_n^5}{(1)_n^5}\frac{(-1)^n}{4^n}(20n^2+8n+1),\qquad T(N)=\frac{\left(\frac12\right)_N^5}{(1)_N^5}\frac{(-1)^N}{4^N}, \tag{19}$$

We have the following p-adic expansions:

$$S(\nu p)=S(\nu)p^2+T(\nu)\left\{7\cdot 2^6\zeta_p(3)\nu^5p^5+B_p\nu^6p^6+C_p\nu^7p^7+\cdots\right\}, \tag{20}$$

We can combine S_p and S_{2p} for eliminating the first term, and we get

$$S(p)-\frac{128}{99}S(2p)=\mathcal{O}(p^5), \tag{21}$$

In the same way, we can combine S_p and S_{2p} for eliminating the term with $\zeta_p(3)$, and we obtain

$$\frac{1701}{256}S(p)+\frac72 S(2p)=\frac{1197}{128}p^2+\mathcal{O}(p^6), \tag{22}$$

and we can combine them for eliminating the term with B_p, and we obtain

$$972S(p) + 256S(2p) = 1170p^2 - 1701\zeta_p(3)p^5 + \mathcal{O}(p^7). \tag{23}$$

$$\frac{1767133000}{19873929}S(p) + \frac{81002624000}{1609788249}S(2p) - \frac{79691776}{1524825}S(3p) - \frac{10200547328}{96994275}S(4p) = \mathcal{O}(p^7). \tag{24}$$

Example 3.2. From

$$\sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}\right)_n \left(\frac{1}{3}\right)_n \left(\frac{2}{3}\right)_n \left(\frac{1}{6}\right)_n \left(\frac{5}{6}\right)_n}{(1)_n^5} (-1)^n \left(\frac{3^6}{4^6}\right)^n (1930n^2 + 549n + 45) = \frac{384}{\pi^2}, \tag{25}$$

we have

$$R(-n, x) = 7 \cdot 2^{14}\,\zeta(3)x^5 + Bx^6 + Cx^7 + \cdots,$$

and the p-adic expansions

$$S(\nu p) = S(\nu)p^2 + 7 \cdot 2^{14}\zeta_p(3)T_5(\nu)p^5 + B_pT_6(\nu)p^6 + C_pT_7(\nu)p^7 + \cdots, \tag{26}$$

where

$$S(N) = \sum_{n=0}^{N-1} \frac{\left(\frac{1}{2}\right)_n \left(\frac{1}{3}\right)_n \left(\frac{2}{3}\right)_n \left(\frac{1}{6}\right)_n \left(\frac{5}{6}\right)_n}{(1)_n^5} \left(-\frac{3^6}{4^6}\right)^n (1930n^2 + 549n + 45),$$

and

$$T(N) = \frac{\left(\frac{1}{2}\right)_N \left(\frac{1}{3}\right)_N \left(\frac{2}{3}\right)_N \left(\frac{1}{6}\right)_N \left(\frac{5}{6}\right)_N}{(1)_N^5} \left(-\frac{3^6}{4^6}\right)^N.$$

Example 3.3. From

$$\sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}\right)_n \left(\frac{1}{3}\right)_n \left(\frac{2}{3}\right)_n \left(\frac{1}{6}\right)_n \left(\frac{5}{6}\right)_n}{(1)_n^5} (-1)^n \frac{5418n^2 + 693n + 29}{80^{3n}} = \frac{128\sqrt{5}}{\pi^2},$$

we have

$$R(-x, x) = 42000L_5(3)x^5 + Bx^6 + Cx^7 + \cdots,$$

and the following p-adic identities

$$S(\nu p) = \left(\frac{5}{p}\right) S(\nu)p^2 + T(\nu)\left\{42000L_{5,p}\nu^5p^5 + B_p\nu^6p^6 + C_p\nu^7p^7 + \cdots\right\}, \tag{27}$$

where

$$S(N) = \sum_{n=0}^{N-1} \frac{\left(\frac{1}{2}\right)_n \left(\frac{1}{3}\right)_n \left(\frac{2}{3}\right)_n \left(\frac{1}{6}\right)_n \left(\frac{5}{6}\right)_n}{(1)_n^5} \left(-\frac{1}{80^3}\right)^n (5418n^2 + 693n + 29),$$

and

$$T(N) = \frac{\left(\frac{1}{2}\right)_N \left(\frac{1}{3}\right)_N \left(\frac{2}{3}\right)_N \left(\frac{1}{6}\right)_N \left(\frac{5}{6}\right)_N}{(1)_N^5} \left(-\frac{1}{80^3}\right)^N.$$

Example 3.4. From

$$\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^{3(n+x)}\frac{\left(\frac{1}{2}\right)_{n+x}^3\left(\frac{1}{3}\right)_{n+x}\left(\frac{2}{3}\right)_{n+x}}{(1)_{n+x}^5}[74(n+x)^2+27(n+x)+3]=\frac{48}{\pi^2},$$

we have

$$R(-n,x)=-7\cdot 2^8\zeta(3)x^5+Bx^6+Cx^7+\cdots+\cdots,$$

and p-adic identities like

$$\sum_{n=0}^{p-1}\left(\frac{3}{4}\right)^{3n}\frac{\left(\frac{1}{2}\right)_n^3\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n}{(1)_{n+x}^5}(74n^2+27n+3)=3p^2+21\zeta_p(3)p^5+\cdots.$$

and

$$\sum_{n=0}^{2p-1}\left(\frac{3}{4}\right)^{3n}\frac{\left(\frac{1}{2}\right)_n^3\left(\frac{1}{3}\right)_n\left(\frac{2}{3}\right)_n}{(1)_{n+x}^5}(74n^2+27n+3)=\frac{135}{32}p^2+\frac{8505}{128}\zeta_p(3)p^5+\cdots.$$

Example 3.5. From

$$\sum_{n=0}^{\infty}\frac{\left(\frac{1}{2}\right)_n^7\left(\frac{1}{4}\right)_n\left(\frac{3}{4}\right)_n}{(1)_n^9}\frac{43680n^4+20632n^3+4340n^2+466n+21}{4096^n}=\frac{2048}{\pi^4}$$

we have

$$R(-n,x)=-95232\zeta(5)x^9+Bx^{10}+\cdots,$$

and the following p-adic expansions:

$$S(\nu p)=S(\nu)p^4+T(\nu)\left\{-95232\zeta_p(5)\nu^9p^9+B_p\nu^{10}p^{10}+\cdots\right\},\tag{28}$$

Example 3.6. The following Ramnujan-like series was discovered by Boris Gourevitch [1]:

$$\sum_{n=0}^{\infty}\frac{\left(\frac{1}{2}\right)_n^7}{(1)_n^7}\left(\frac{1}{64}\right)^n(168n^3+76n^2+14n+1)=\frac{32}{\pi^3}.\tag{29}$$

We have

$$R(-n,x)=1536L_{-4}(4)\,x^7+Bx^8+Cx^9+\cdots,\tag{30}$$

and the following p-adic identities:

$$S(\nu p)=\left(\frac{-4}{p}\right)S(\nu)p^3+T(\nu)\left\{1536L_{-4,p}(4)\nu^7p^7+B_P\nu^8p^8+C_p\nu^9p^9+\cdots\right\}.$$

4. Proofs of rational Ramanujan-like series for π^{-m} via bilateral series and conjectured p-adic identities

Take a rational value z_0 and a character χ_0, and substitute x in (12) with $3m+1$ values such that the difference between any two of them is not an integer (for avoiding periodicity), then we obtain a linear system of $3m+1$ equations with $m+1$ unknowns a_k, m unknowns α_k, and m unknowns β_k. Hence, $3m+1$ values of x determine the parameters

a_k, α_k, and β_k, and the solution is unique. If we take the $3m+1$ values of x very small, then we can ignore the summation

$$R(-n,x) = x^{2m+1}\left(\prod_{i=0}^{2m}\frac{(1)_{n-x}}{(s_i)_{n-x}}\right)\sum_{k=0}^{m} a_k(-n+x)^{k-2m-1}\, z_0^{-n} = \mathcal{O}(x^{2m+1}), \qquad (31)$$

as it is very close to 0, and it tends to 0 taking sequences of values of x tending to 0. However, we can only obtain approximations of the parameters. We can write a Maple program for evaluating the bilateral sum (12) at $x = x_0$.

MAPLE PROGRAM (case m=2):

```
restart: alias(po=pochhammer): with(numtheory):

bilater:=proc(s1,s2,z0,x,j)
local p,s0,s3,s4,z1,h,hh,r,u,v;
p:=(a,b)->pochhammer(a,b): s0:=1/2; s3:=1-s1; s4:=1-s2:
h:=n->p(s0,n)*p(s1,n)*p(s2,n)*p(s3,n)*p(s4,n)/p(1,n)^5:
if type(x,integer) then; hh:=n->h(n)*n^j:
r:=evalf((-1)^x*subs(z=z0,sum(hh(n)*z^n,n=0..infinity))):
return r; else;
if z0<0 then;
z1:=z0; hh:=(j,n)->(-1)^n*h(n+x)*(n+x)^j;
else
z1:=-z0; hh:=(j,n)->exp(-I*Pi*x)*h(n+x)*(n+x)^j;
fi;
assume(zz<1);
u:=j->convert(Sum(hh(j,n)*(-zz)^(n+x),n=0..infinity),hypergeom):
assume(z>1);
v:=j->convert(Sum(hh(j,-n)*(-z)^(-n+x),n=1..infinity),hypergeom):
r:=subs({z=z1,zz=z1},u(j)+left*v(j)): return r: fi; end:

ramabilater:=proc()
global f,eq,sol,a,alpha,aa0,aa1,aa2,alfa1,alfa2;
f:=x->1/Pi^2*cos(Pi*x)*(cos(Pi*x)-cos(Pi*s1))/(1-cos(Pi*s1))*
(cos(Pi*x)-cos(Pi*s2))/(1-cos(Pi*s2))*
(cos(Pi*x)-cos(Pi*s3))/(1-cos(Pi*s3))*
(cos(Pi*x)-cos(Pi*s4))/(1-cos(Pi*s4))*
(1+alpha1*(cos(2*Pi*x)-1)+alpha2*(cos(4*Pi*x)-1)):
eq:=x->a0*evalf(bilater(s1,s2,z0,x,0))+
a1*evalf(bilater(s1,s2,z0,x,1))+
a2*evalf(bilater(s1,s2,z0,x,2))-evalf(f(x)):
sol:=solve({eq(x1),eq(x2),eq(x3),eq(x4),eq(x5)},
{alpha1,alpha2,a0,a1,a2}):
aa0:=evalf(subs(sol,a0),100); aa1:=evalf(subs(sol,a1),100);
aa2:=evalf(subs(sol,a2),100);
alfa1:=evalf(subs(sol,alpha1),100);
```

```
alfa2:=evalf(subs(sol,alpha2),100);
print(s[0]=1/2,s[1]=s1,s[2]=s2,s[3]=1-s1,s[4]=1-s2,z[0]=z0): print(LEFT‘=left):
print(a[0]=aa0,a[1]=aa1,a[2]=aa2,alpha[1]=alfa1,alpha[2]=alfa2);
end:
```

The procedure `bilater`; calculates the bilateral sum

$$\sum_{n\in\mathbb{Z}}(-1)^n\left[\prod_{i=0}^{4}\frac{(s_i)_{n+x}}{(1)_{n+x}}\right](-z)^{n+x}\,(n+x)^j.$$

The procedures `bilater`; and `ramabilater`; help the reader to check the examples. If $x=\varepsilon_0$ and `left` $=1$ then it means that we are taking into account the summation to the left side:

$$\sum_{n=1}^{\infty}R(-n,x)=x^{2m+1}\sum_{n=1}^{\infty}\left(\prod_{i=0}^{2m}\frac{(1)_{n-x}}{(s_i)_{n-x}}\right)\sum_{k=0}^{m}a_k(-n+x)^{k-2m-1}\,z_0^{-n}, \tag{32}$$

and if `left` $=0$ the program does not take it into account. In the p-adic system, the zeros are the primes because $p\equiv 0 \pmod p$. Hence, $3m+1$ primes are enough to determine the parameters a_k from the supercongruences

$$S(p)\equiv\left(\frac{\chi_0}{p}\right)S(1)p^m \pmod{p^{2m+1}}. \tag{33}$$

Example 4.1.
```
Digits:=200:
s1:=1/2: s2:=1/2: z0:=-1/2^(10): left:=0:
x1:=sqrt(2)*10^(-40): x2:=sqrt(3)*10^(-40):
x3:=sqrt(5)*10^(-40): x4:=sqrt(6)*10^(-40):
x5:=sqrt(7)*10^(-40):

ramabilater();
```

Example 4.2.
```
Digits:=200:
s1:=1/3: s2:=1/4: z0:=-1/48: left:=0:
x1:=sqrt(2)*10^(-40): x2:=sqrt(3)*10^(-40):
x3:=sqrt(5)*10^(-40): x4:=sqrt(6)*10^(-40):
x5:=sqrt(7)*10^(-40):

ramabilater();
```

5. Conclusion

The conjecture (17), implies the following supercongruences (conjectured in [7] by Wadim Zudilin):

$$S(p)\equiv\left(\frac{\chi_0}{p}\right)S(1)p^m \pmod{p^{2m+1}}. \tag{34}$$

Let z_0 be a rational number and χ_0 a character. Substituting p with $3m+1$ prime numbers we get a system of supercongruences $\pmod{p^{2m+1}}$. This system has no solution

or a unique solution (but rational multiples): a_k/t (t rational), and there exists a rational number $t = t_0$ such that

$$\sum_{n=0}^{\infty}\left(\prod_{i=0}^{2m}\frac{(s_i)_n}{(1)_n}\right)\sum_{k=0}^{m}a_k n^k z_0^n = t_0\frac{\sqrt{d}}{\pi^m}. \tag{35}$$

That is, if for a rational z_0 and a character $\chi_0 = d$, a system of supercongruences (34) has a unique solution (but rational multiples t), then it is the unique solution of a rational Ramanujan-like series for π^{-m} (except for a rational multiple t_0).

Example 5.1.

$$\sum_{n=0}^{\infty}\frac{\left(\frac12\right)_n\left(\frac13\right)_n\left(\frac23\right)_n\left(\frac14\right)_n\left(\frac34\right)_n}{(1)_n^5}\frac{(-1)^n}{48^n}(a_2n^2+a_1n+a_0) = \frac{48}{\pi^2}. \tag{36}$$

MAPLE PROGRAM (system of congruences):

```
restart: alias(po=pochhammer): with(numtheory):
A:=(a,b,c,n)->po(1/2,n)*po(1/3,n)*po(2/3,n)*po(1/4,n)*po(3/4,n)
/po(1,n)^5*(c*n^2+b*n+a)*(-1/48)^n;
v:=(a,b,c,p)->sum(A(a,b,c,n),n=0..p-1);
y:=(a,b,c,p)->v(a,b,c,p)-jacobi(1,p)*A(a,b,c,0)*p^2 mod p^5;

SEARCH:=proc(); global a,b,c;
for a from -15 to 15 do; for b from -200 to 200 do;
for c from -800 to 800 do;
if y(a,b,c,5) = 0 and y(a,b,c,11)=0 and y(a,b,c,13) = 0
and y(a,b,c,17) = 0 and y(a,b,c,19)=0 and y(a,b,c,23) = 0
and y(a,b,c,29) = 0 then print(a,b,c); end if:
end do; end do; end do; end:

SEARCH();
```

NOTE (number of supercongruences). As all the β_k are equal to 0, it seems that we can reduce the number of necessary supercongruences to $2m+1$.

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References

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